

Transfer Learning for Binary Food Image Classification

INDENG 1/242B Spring 2026 Project

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May 2026

Motivation and Learning Problem

Use case

Coarse food-image screening: classify a food image as healthy or unhealthy.

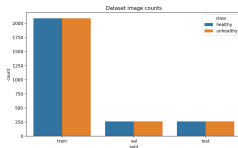
- Input feature: RGB food image x .
- Label: $y \in \{\text{healthy}, \text{unhealthy}\}$.
- Learned mapping: $f_{\theta}(x) \rightarrow y$.
- Goal: compare a task-specific CNN with transfer learning and fine-tuning.

Project questions

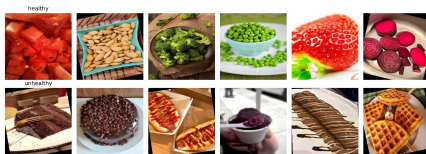
- How strong is a small CNN baseline?
- How much does ImageNet transfer help?
- Does fine-tuning improve the transferred model?
- What errors remain?

Dataset and Preprocessing

Split	H	U	Total
Train	2089	2089	4178
Val.	261	261	522
Test	261	261	522



- 5,222 total images.
- Balanced classes in every split.
- Images resized to 128×128 .
- Training: crop, flip, color jitter.
- Evaluation: resize, center crop, ImageNet normalization.



Methodology

Models

- Simple CNN: Conv–BatchNorm–ReLU blocks, max pooling, adaptive average pooling, dropout, linear head.
- Frozen ResNet18: ImageNet encoder fixed, binary head trained.
- Fine-tuned ResNet18: layer4-only and full-backbone variants.
- Advanced extension: supervised contrastive pretraining.

Classifier probabilities:

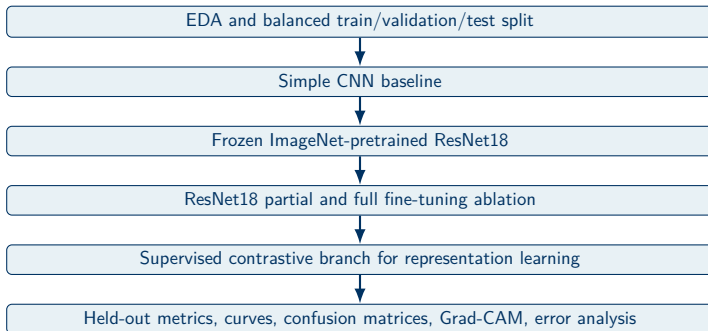
$$p_{\theta}(y = c \mid x) = \frac{\exp z_c}{\sum_{k=1}^2 \exp z_k}.$$

Regularized objective:

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N L_{\text{ce}}(f_{\theta}(x_i), y_i) + \lambda \|\theta_{\text{train}}\|_2^2.$$

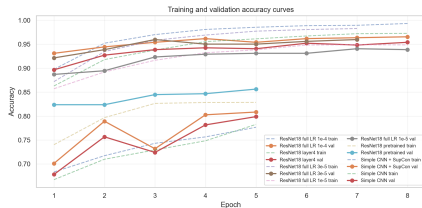
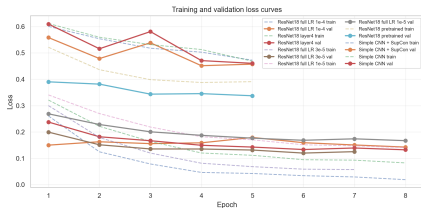
- Optimizer: AdamW.
- Regularization: weight decay, dropout, augmentation, early stopping.

Experimental Pipeline



The pipeline covers problem formulation, network design, objective, optimizer, regularization, advanced method, and evaluation diagnostics.

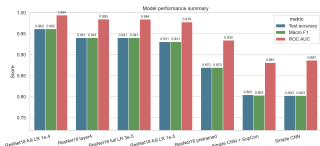
Training and Validation Diagnostics



- Simple CNN variants underfit relative to the transferred ResNet18 family.
- Best fine-tuned ResNet18 keeps improving validation accuracy through the final epoch.
- A larger train-validation gap appears during fine-tuning, so validation monitoring is necessary.

Main Results

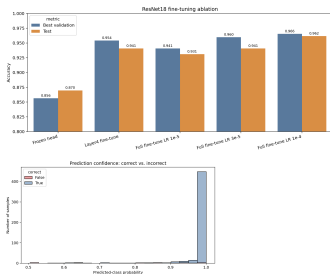
Experiment	Val.	Test	F1	AUC
ResNet18 full fine-tune, LR 10^{-4}	0.9655	0.9617	0.9617	0.9944
ResNet18 layer4 fine-tune, LR 3×10^{-5}	0.9540	0.9406	0.9406	0.9850
ResNet18 full fine-tune, LR 3×10^{-5}	0.9598	0.9406	0.9406	0.9845
ResNet18 frozen backbone	0.8563	0.8697	0.8697	0.9344
Simple CNN + SupCon fine-tune	0.8084	0.8046	0.8033	0.8813
Simple CNN baseline	0.7989	0.8027	0.8027	0.8869



Practical improvement

Full fine-tuning improves test accuracy from 0.8697 to 0.9617: a 9.20-point absolute gain over frozen transfer.

Ablation and Error Analysis



Ablation finding

- Frozen head: 0.8697 test accuracy.
- Layer4 fine-tune: 0.9406.
- Full fine-tune at 10^{-4} : 0.9617.

Residual errors

- 20 errors out of 522 test images.
- 10 healthy→unhealthy and 10 unhealthy→healthy.
- 12 high-confidence errors with confidence ≥ 0.80 .

Safety, Ethics, and Takeaways

Deployment risks

- healthy / unhealthy labels are coarse nutritional proxies.
- Food meaning varies by portion size, ingredients, cuisine, and medical context.
- A high-accuracy model still makes high-confidence mistakes.
- User-uploaded images raise privacy and consent concerns.

Conclusion

- Transfer learning is the dominant performance driver.
- Best model: 0.9617 accuracy and 0.9944 ROC AUC.
- Current system is suitable as an academic classifier, not an authoritative health tool.
- Next steps: calibration, larger backbones, external validation.

Thank You

Interactive demo: <https://huggingface.co/spaces/LittleOtterUD/INDENGInterface>